

1 Summary

Model (Run-name): model1 (test)

Iteration: 27

TAC: 154882.8472463531

Took: 2.6816470623016357s

Total Time: 49.45157551765442s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

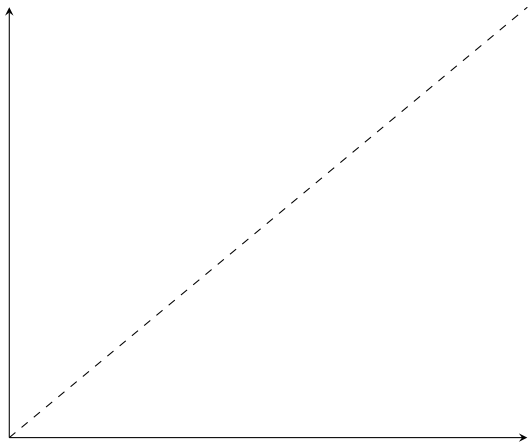
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

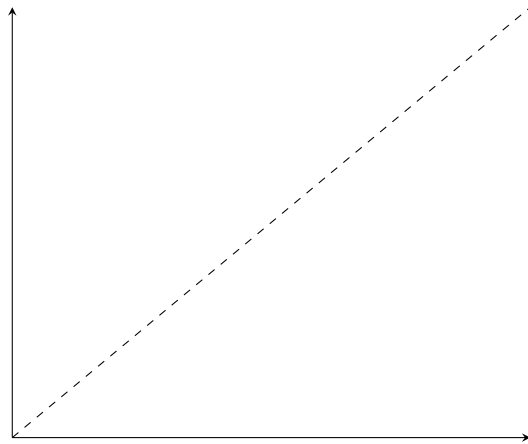
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05168374363567882	$b_{1,1,1}^{H,out}$	921.3705826663775
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.018272741491861907	$b_{1,2,2}^{H,out}$	3431.144856620122
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02767625751785106	$b_{2,1,2}^{H,out}$	9395.880120864536
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014790697383470075	$b_{1,1,1}^{C,in}$	4982.893027340363
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009533205393224126	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022434651221490052	$b_{2,1,2}^{C,in}$	4116.451687227616
$A_{1,1,1}^\beta$	72.01156079815767	$q_{1,1,1}$	697.2534139096911	$b_{1,1,1}^{C,out}$	5680.146441250054
$A_{1,2,2}^\beta$	71.26369181826138	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	132.5928424015674	$q_{2,1,2}$	2404.1198791354636	$b_{2,1,2}^{C,out}$	6520.571566363081
$A_{cu,1}^\beta$	4.440201431981878	$q_{cu,1}$	152.74658609030894		
$A_{cu,2}^\beta$	38.035748486883605	$q_{cu,2}$	1995.8801208645364		
$A_{hu,1}^\beta$	13.411355634100326	$q_{hu,1}$	498.6267069548451		
$A_{1,1,1}$	72.01156079815767	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	71.26369181826138	$t_{1,2}^{(H)}$	580.2746586090309		
$A_{2,1,2}$	132.5928424015674	$t_{1,3}^{(H)}$	385.2746586090309		
$A_{cu,1}$	4.440201431981878	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	38.035748486883605	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.411355634100326	$t_{2,3}^{(H)}$	469.79400604322683		
$dt_{1,1,1}$	33.24178046365634	$t_{1,1}^{(C)}$	616.7582195363436		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	570.2746586090309		
$dt_{1,2,2}$	80.2746586090309	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	35.274658609030894	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	19.725341390969106	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	59.794006043226815	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	65.2746586090309	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	149.79400604322683	$t_{1,2,2}^H$	370.0724191628162		
$dt_{hu,1}$	63.24178046365634	$t_{2,1,2}^H$	469.79400604322683		
$f_{1,1,1}^H$	2.4901907639631826	$t_{1,1,1}^C$	646.1955328332481		
$f_{1,2,2}^H$	9.27336447735168	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	649.4511645803842		
$f_{1,1,1}^C$	8.738686832692391	$b_{1,1,1}^{H,in}$	1618.6239965760687		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5381.1448566201225		
$f_{2,1,2}^C$	10.04012606640882	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

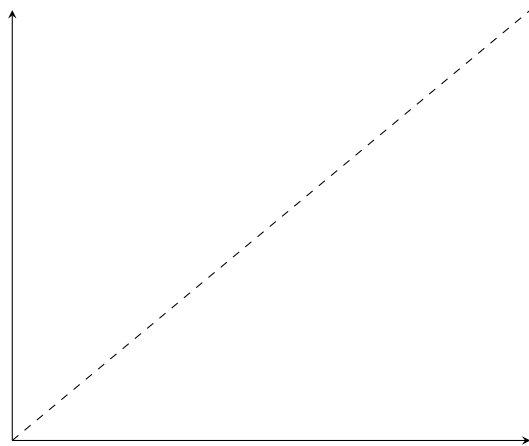
- $(1, 1, 1)$
 - Value: 0.051684
 - Error: 0.000000009307675
 - Relative error: 0.000000180088996
- $(1, 2, 2)$
 - Value: 0.018273
 - Error: 0.000000351119085
 - Relative error: 0.000019215088161
- $(2, 1, 2)$
 - Value: 0.027676
 - Error: 0.000001262011626
 - Relative error: 0.000045596991603
- $cu, 1$
 - Value: 0.014791
 - Error: 0.000000055651448
 - Relative error: 0.000003762583802
- $cu, 2$
 - Value: 0.009533
 - Error: 0.000000916941209
 - Relative error: 0.000096174684689
- $hu, 1$
 - Value: 0.022435
 - Error: 0.000000001012990
 - Relative error: 0.000000045152913



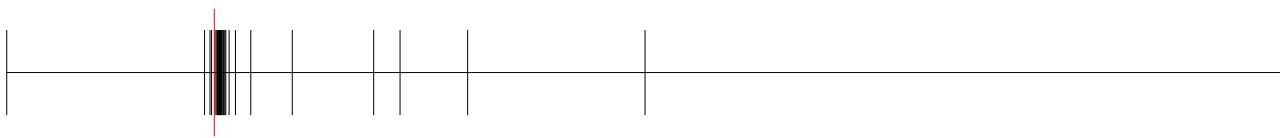
(a) *RecLMTD* - (1, 1, 1)



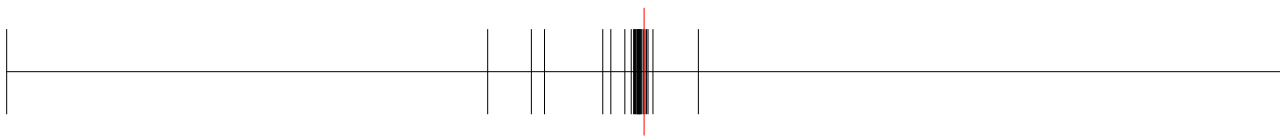
(b) *RecLMTD* - (1, 2, 2)



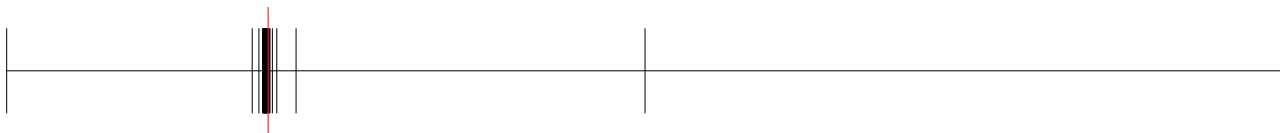
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1618.6239965760687
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5381.1448566201225
 - Absolute error: 0.046450367761281086
 - Relative error: 8.632134975883336e-06
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 921.3705826663775
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.2338882948988323e-16
- (1, 2, 2)
 - Value: 3431.144856620122
 - Absolute error: 0.6715692919387948
 - Relative error: -0.00019568916532600205
- (2, 1, 2)
 - Value: 9395.880120864536
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 4982.893027340363
 - Absolute error: 0.5586228645233859
 - Relative error: -0.00011209557225270142
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4116.451687227616
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 5680.146441250054
 - Absolute error: 33.24604713550525
 - Relative error: 0.005887486021562513
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6520.571566363081
 - Absolute error: 0.0
 - Relative error: 0.0



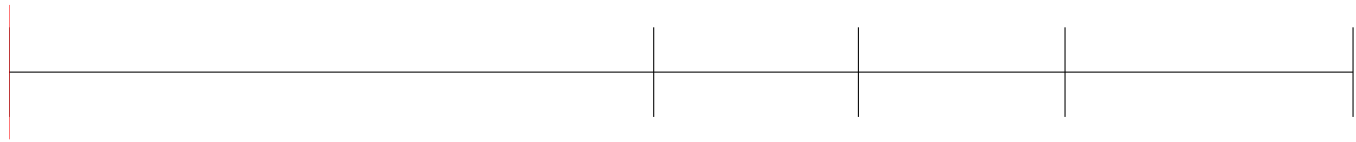
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



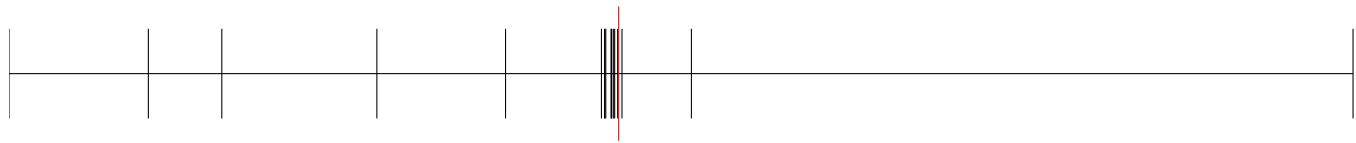
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



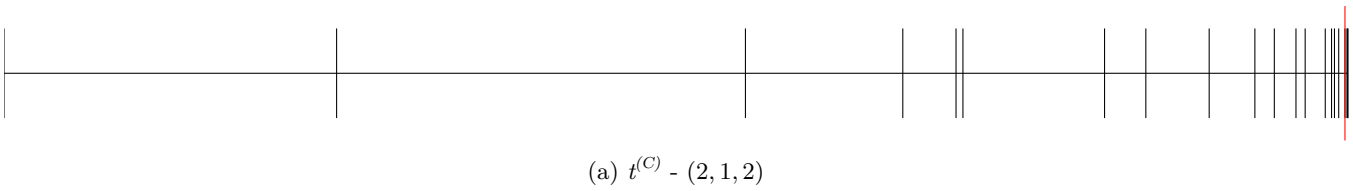
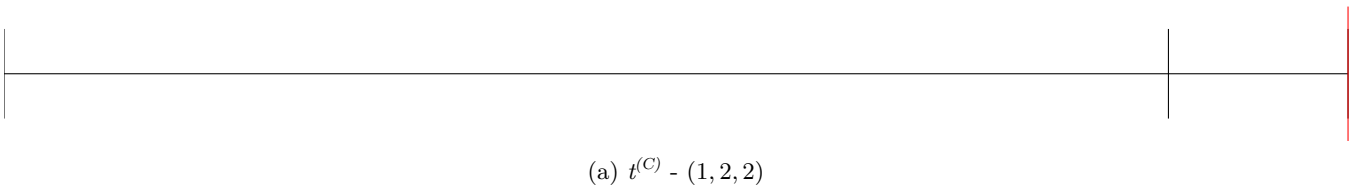
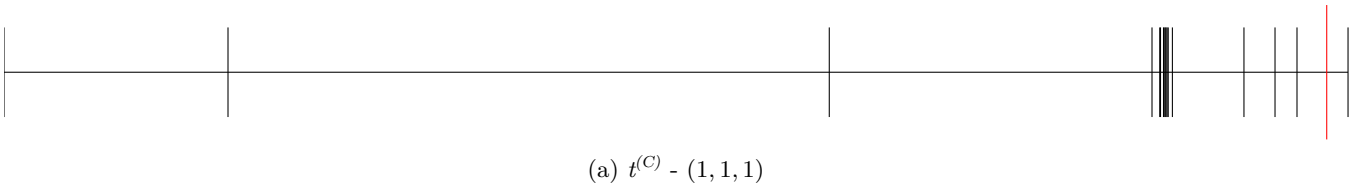
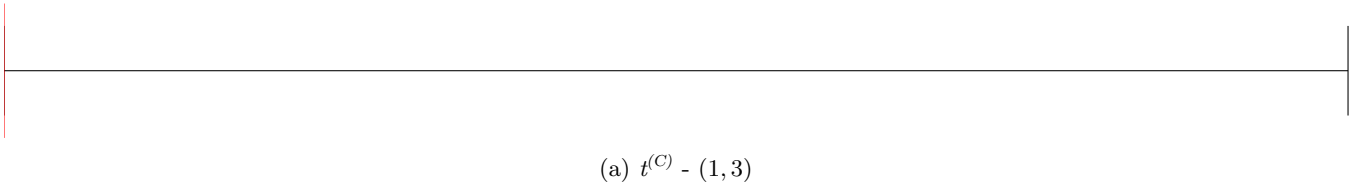
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

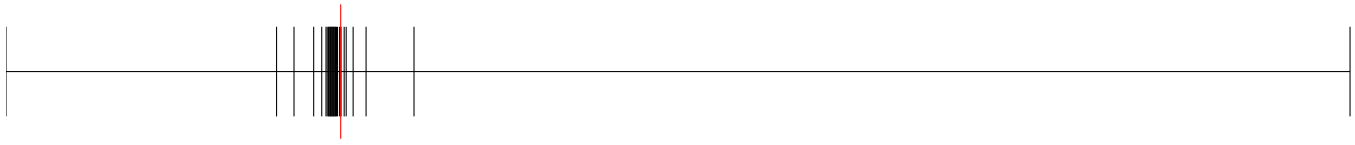
- $(1, 1, 1)$
 - Value: 72.011561
 - Error: 0.061772589062969
 - Relative error: 0.000857079673714
- $(1, 2, 2)$
 - Value: 71.263692
 - Error: 0.000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 132.592842
 - Error: 0.481239355908713
 - Relative error: 0.003616326707298

5.2 A_{CU}

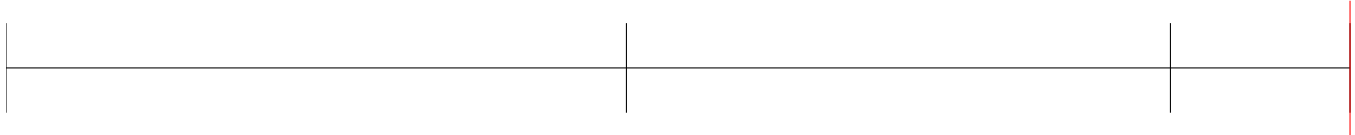
- $cu, 1$
 - Value: 4.440201
 - Error: 0.078255630457960
 - Relative error: 0.017319104591802
- $cu, 2$
 - Value: 38.035748
 - Error: 0.018521778025637
 - Relative error: 0.000486720094662

5.3 A_{HU}

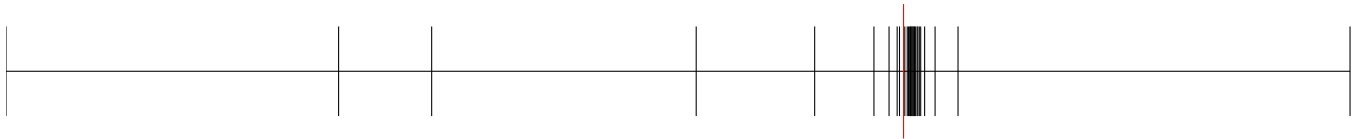
- $hu, 1$
 - Value: 13.411356
 - Error: 0.012463878202166
 - Relative error: 0.000928489703750



(a) $A^\beta - (1, 1, 1)$



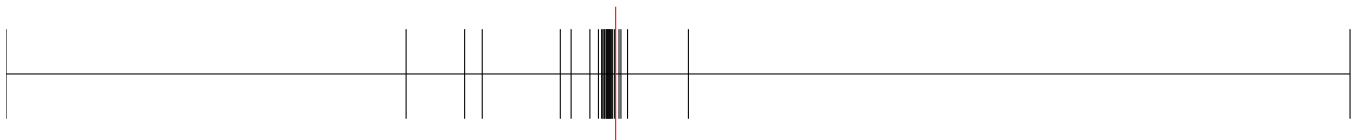
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$



(a) $A^\beta - hu, 1$

6 A^β Breakpoints

- (1,1,1)
 - q : 72.011561
 - $\hat{q}^\beta$: 72.011561
 - q^β : 72.011561
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, **71.88123217857378**, **72.36161285231599**, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,2,2)
 - q : 71.263692
 - $\hat{q}^\beta$: 71.263692
 - q^β : 71.263692
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, **70.94081287188642**, **72.25348351595485**, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 132.592842
 - $\hat{q}^\beta$: 132.592842
 - q^β : 132.592842
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, **132.12228195956155**, **135.81963163113545**, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.440201
 - $\hat{q}^\beta$: 4.440201
 - q^β : 4.440201
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, **4.432280503220532**, **4.613047202566896**, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335, 4.974201802143407, 4.983938626310074,

4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.195781532076625,
7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $cu, 2$

– q : 38.035748

– $\hat{q}^\beta$: 38.035748

– q^β : 38.035748

– Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717,
36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.52720514437563,
37.554741031910254, 37.612770977560544, 37.65435035996605, 37.6551299822246, 37.682669717644174,
37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985,
37.9454585315849, **38.10872593446926**, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $hu, 1$

– q : 13.411356

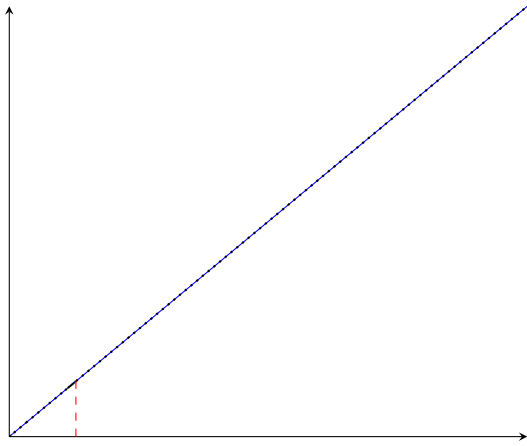
– $\hat{q}^\beta$: 13.411356

– q^β : 13.411356

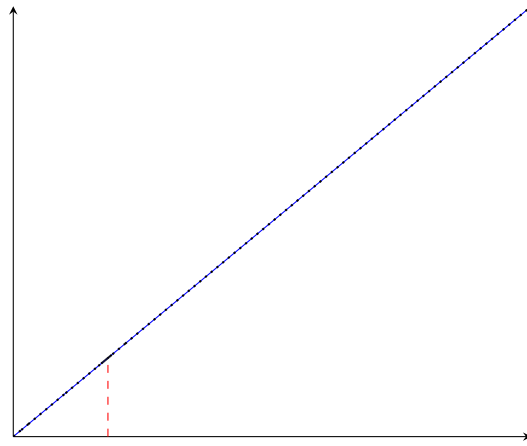
– Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861,
13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.211215176722128,
13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171,
13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218,
13.32663103908035, **13.346967780023082**, **13.439500729482463**, 13.468285806082731, 13.498843422507207,
14.09178614751222, 237.3002543523117]

– Error: 0.0000000000000000

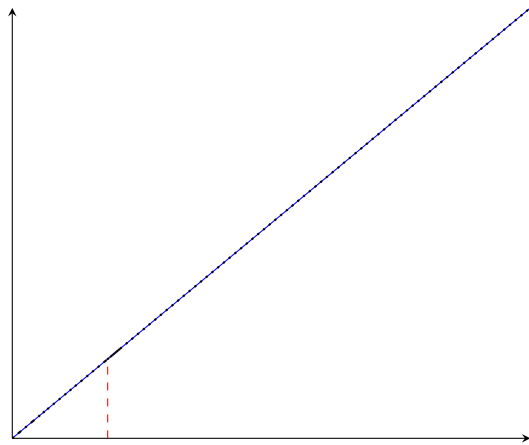
– Relative error: 0.0000000000000000



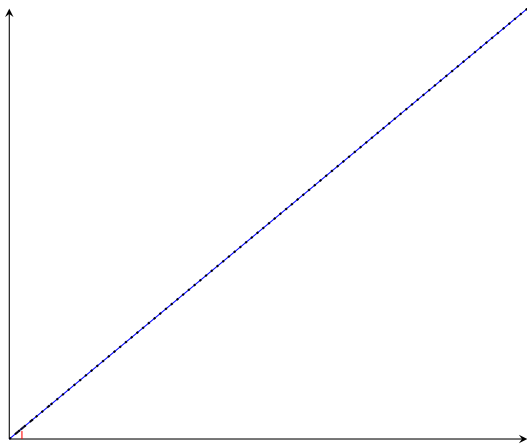
(a) $q^\beta - (1, 1, 1)$



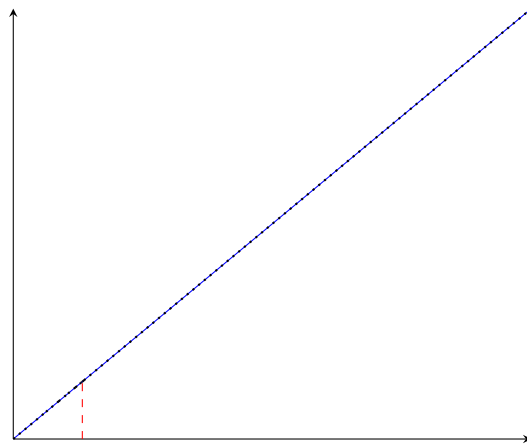
(b) $q^\beta - (1, 2, 2)$



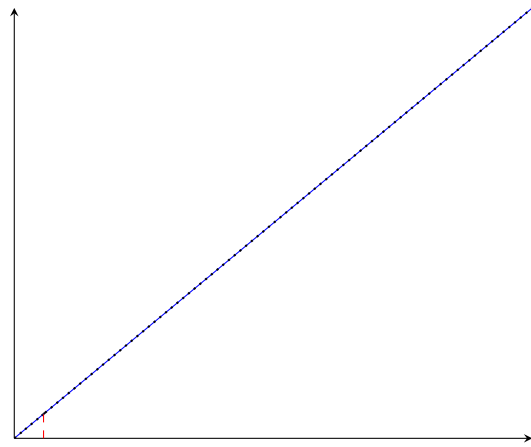
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	2033.5483127723835
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008110829165264705	$b_{2,1,1}^{C,in}$	3571.2268517951015
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038172	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	2033.5483127723828
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	3571.2268517951015
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	564.6191576569319		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	576.8824300983304		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	573.147447436938		
$dt_{2,2,2}$	10.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	350.0		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	421.6017294701868		
$f_{1,1,2}^H$	0.7266355226483187	$b_{1,2,1}^{H,in}$	4881.376003423931		
$f_{1,2,1}^H$	7.509809236036817	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	421.6017294701868		
$f_{1,1,2}^C$	4.9598739335911795	$b_{1,2,1}^{H,out}$	4881.376003423931		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	6.261313167307609	$b_{2,2,1}^{H,out}$	0.0		